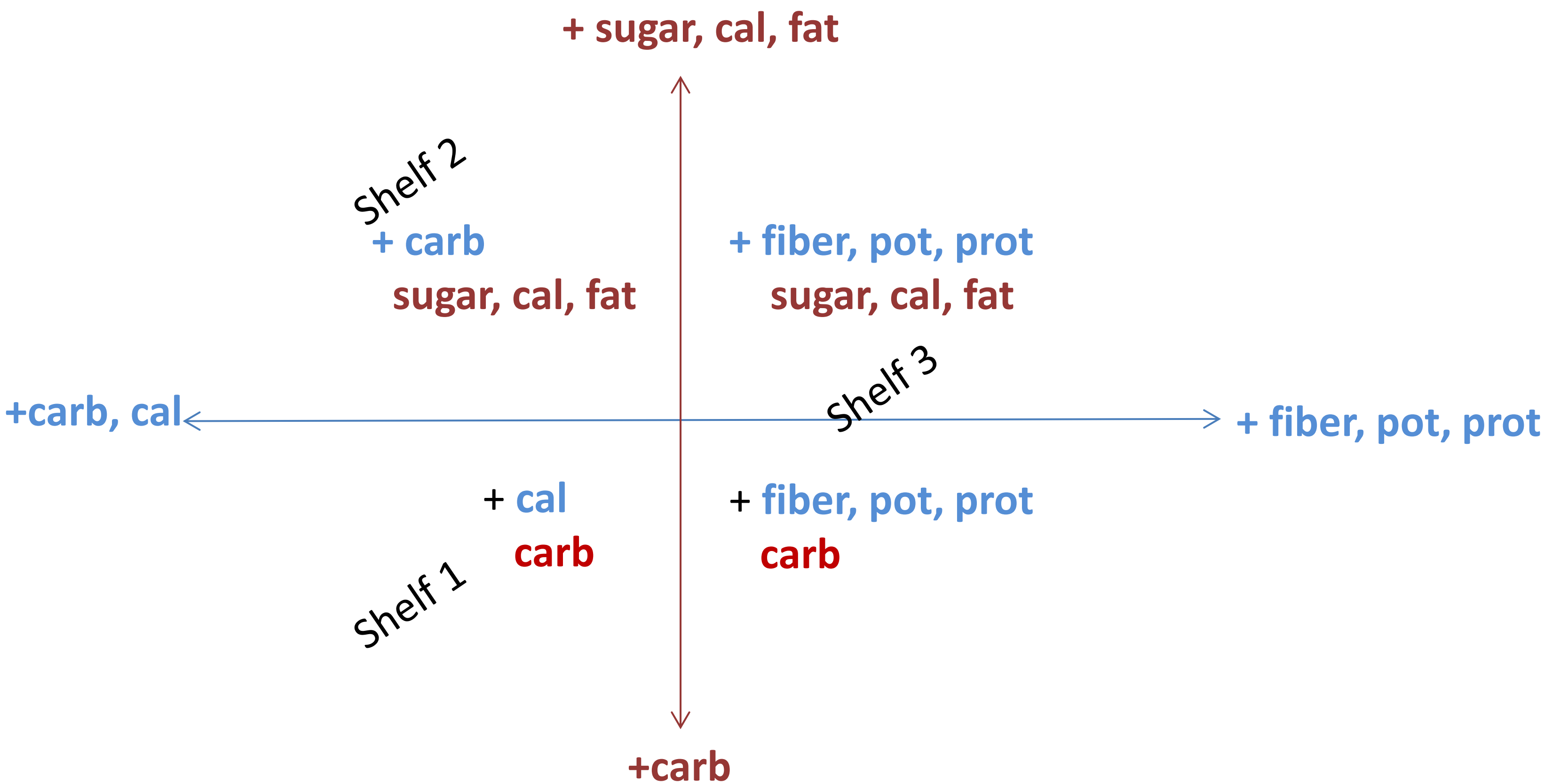
Rough interpretation of the cereals PCA results:

First principal component is a linear combination of fiber, potassium and protein (and with less and negative weight, variables calories and carbohydrates). Not being a nutritionist, cereals with a high value of this first pc can be classified as healthier than the others. The second PC is a linear combination of sugars, calories and fat (and with less and negative weight, variable carbohydrates), a possible indicator of non healthiness. This two first PC account for a 58.39% of the variance in the data set. It seems than healthier cereals are in shelf 3, followed by shelf 1 and shelf 2.

The variables that are best represented by this two dimensional graph are: fiber, potassium, sugars and carbohydrates. Fiber and potassium are also highly correlated. Sodium is the variable with the worst representation on this map.

What are the healthiest cereals? And the less healthy ones?

Rough interpretation of the cereals PCA map:



You have to bear in mind that, altough we have treated protein and fat as quantitative continuous variables, they have few different values. Another analysis may have explored the results using them as categorical. Moreover, due to the low frequency of cereals with proteins equal to 4 and six, we can create a new category “+4” with 7 observations for cereals with 4 or more proteint content.

```
> table(cereal.pc$fat)
 0  1  2  3
22 28 12  3

> table(cereal.pc$prot)
 1  2  3  4  6
13 20 25  5  2
```